DIAGNOSTIC PERFORMANCE OF RADIOLOGIST AND A REAL-TIME COMMERCIAL AI-ASSISTED TOOL BI-RADS CLASSIFICATION OF BIOPSIED BREAST MASSES: A SINGLE-CENTRE PILOT STUDY

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BACKGROUND

Ultrasound interpretation of breast masses using BI-RADS is highly operator-dependent, causing notable interobserver variability. Artificial intelligence (AI)-assisted tool offers a promising solution to enhance diagnostic accuracy. This pilot study compares the diagnostic performance of a real-time commercial AI-assisted ultrasound system with radiologist-assigned BI-RADS classifications, using histopathological examination (HPE) as the reference standard.

METHODS

This prospective, single-centre study evaluated 18 biopsy-proven breast masses collected over a one-month period. Ultrasound images were interpreted by two breast imaging fellows and a commercial AI-assisted tool. According to the BI-RADS lexicon, masses warranting biopsy are classified as BI-RADS 4 or higher; BI-RADS 3 masses typically do not require immediate biopsy. Thus, masses with an inherently low malignancy risk (2–10%) were assigned at least BI-RADS 4a to consider tissue biopsy. Diagnostic performance metrics were compared across two analytical frameworks: an original model (BI-RADS 4a treated as "positive") and an adjusted model (BI-RADS 4a reclassified as "negative").

DISCUSSION

Both radiologists and the AI-assisted tool achieved 100% sensitivity for malignant cases; the AI-assisted tool demonstrated superior initial specificity by correctly classifying most benign masses as BI-RADS 2 instead of BI-RADS 4a. Because clinical caution prompts radiologists to overcall low-risk masses as BI-RADS 4a for tissue biopsy, treating BI-RADS 4a as a statistical "positive" artificially inflates false-positive rates. Reclassifying BI-RADS 4a as "benign" for analysis correctly counted these biopsy-negative cases as true negatives, significantly boosting specificity, accuracy, and PPV for both approaches while highlighting AI's potential to reduce unnecessary biopsies.

This study is limited by a small sample size and short study duration of one month. In addition, the AI-assisted tool required some familiarity to operate effectively, while minor technical issues encountered during its use may have further limited the number of cases included. Nevertheless, a malignancy detection rate of 33.3% was achieved during the study period, which is considered acceptable given the diagnostic nature of the cohort.

CONCLUSION

Both radiologists and the AI-assisted tool maintained excellent sensitivity. However, the commercial AI-assisted tool demonstrated superior specificity by correctly classifying most benign lesions, highlighting its potential clinical value in reducing unnecessary breast biopsies. These preliminary findings support the integration of AI-assisted ultrasound systems as a supportive clinical adjunct. Nonetheless, some training and familiarisation with the system may be required for effective use.

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RESULTS

Variable	HPE		
	Benign n(%)	Malignant n(%)	Total n(%)
Age in mean±SD	52.00(8.264)	50.67(12.420)	51.43(9.827)
Risk factor			
No	9(75.0)	5(83.3)	14(77.8)
Yes	3(25.0)	1(16.7)	4(22.2)
Initial presentation			
Diagnostic	9(75.0)	6(100.0)	15(83.3)
Screening	1(8.3)	0(0.0)	1(5.6)
Short term follow-up	2(16.7)	0(0.0)	2(11.1)
Sex			
Female	12(100.0)	6(100.0)	18(100.0)

Table 1: Descriptive statistics of patients (n=18)

	Sensitivity	Specificity	PPV	NPV	AUC	*P*-value
AI-BI-RADS	100.0%	75.0%	66.7%	100.0%	0.875	<0.001
Adjusted AI-BI-RADS	100.0%	91.7%	85.7%	100.0%	0.958	<0.001
Radiologist BI-RADS	100.0%	0.0%	33.3%	-	0.500	>0.95
Adjusted Radiologist BI-RADS	100.0%	100.0%	100.0%	100.0%	1.000	<0.001

Table 2: Comparison of diagnostic accuracy between AI-assisted tool and radiologist BI-RADS classifications

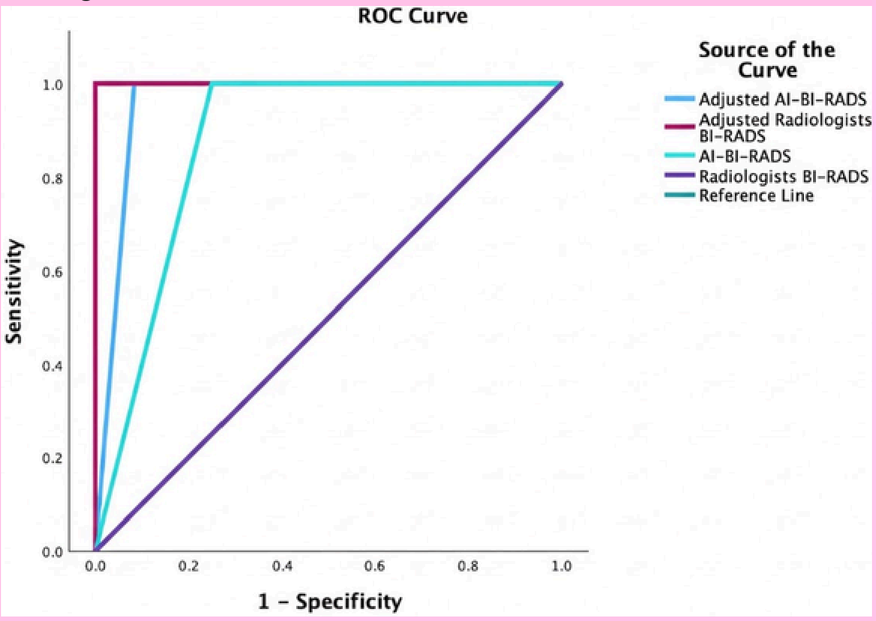

Figure 1: ROC curve showing the diagnostic performance

Key numerical findings

Of the 18 masses, 6 (33.3%) were malignant and 12 (66.7%) were benign. All malignant masses were correctly identified by both radiologists and AI-assisted tool (assigned BI-RADS 4c or 5). For benign masses, radiologists assigned BI-RADS 4a to all 12 cases, whereas the AI-assisted tool classified 9 as BI-RADS 2, 2 as BI-RADS 4a, and 1 as BI-RADS 4c. In the original analysis, radiologists achieved 100% sensitivity and 0% specificity (AUC = 0.500), while the AI-assisted tool demonstrated 100% sensitivity and 75.0% specificity (AUC = 0.875). In the adjusted framework, the radiologists' metrics shifted to 100% sensitivity and 100% specificity (AUC = 1.000), while the AI-assisted tool achieved 100% sensitivity and 97.1% specificity (AUC = 0.958).

FIGURES

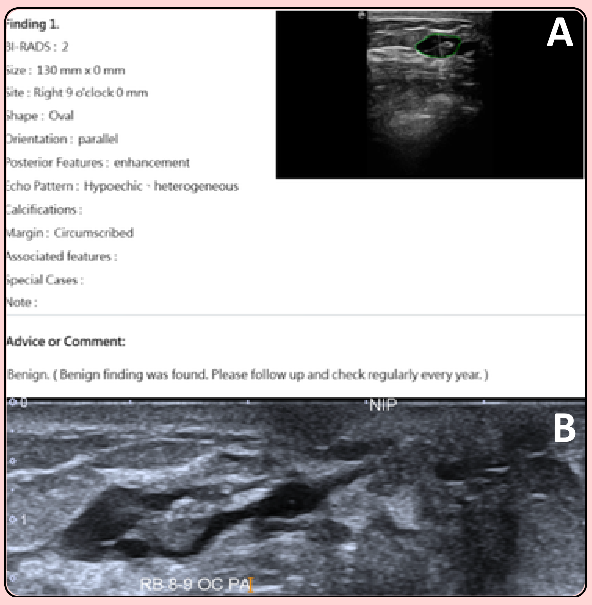

Figure 2:
(A) Commercial AI-assisted tool analysis of the targeted ultrasound (assigned BI-RADS 2).
(B) Corresponding ultrasound image demonstrating a prominent duct with internal echogenicity along the right 8-9 o'clock radian (assigned BI-RADS 4a). HPE confirmed intraductal papilloma with focal usual ductal hyperplasia and apocrine metaplasia.

Figure 3:
(C) Commercial AI-assisted tool analysis of the targeted ultrasound (assigned BI-RADS 5).
(D) Corresponding ultrasound image demonstrating an irregular hypoechoic mass with indistinct margin along the left 2 o'clock radian (assigned BI-RADS 5). HPE confirmed invasive carcinoma.